ATmega328 Analogue I/O Port Expander Using I2C Bus

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This project explains the interfacing of PCF8591 converter (slave) with ATmega328 microcontroller (master) using I2C bus to get direct multiple analogue inputs and outputs. ATmega328 microcontroller (MCU) on the Arduino Uno board consists of only six analogue inputs, which are not enough for some projects that need more analogue inputs. This circuit provides twenty analogue inputs, five analogue outputs and an LCD for display.

Circuit and working

The circuit diagram of ATmega328 analogue I/O port expander using I2C bus is shown in Fig. 1. It is built around five PCF8591 (IC1 through IC5), ATmega328 (IC6), and a few other components.

PCF8591 is basically an 8-bit analogue-to-digital conversion (ADC)

Slave Address Byte — Table 1

	Slave address							
Bit	7 MSB	6	5	4	3	2	1	0 LSB
Slave address	1	0	0	1	A2	A1	A0	R/$\overline{W}$

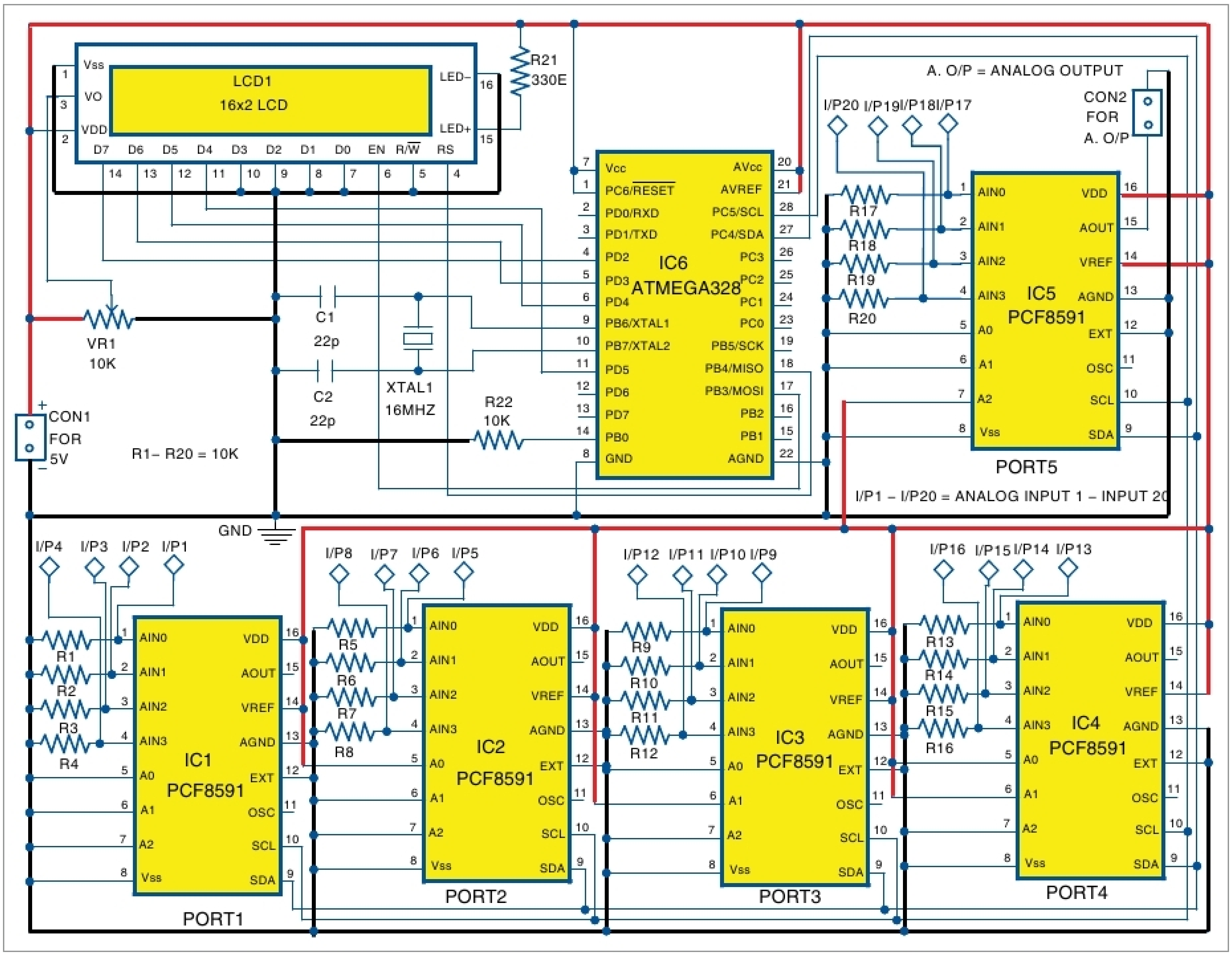

Fig. 1: Circuit diagram of ATmega328 I/O port expander